

Radiosity methods for light diffusion

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Environmental light found in natural daylight and complex artificially lit interiors is one of the most difficult phenomena to model predictively with computational methods. With light diffusing and scattering throughout a scene, only methods that attempt to account for global illumination will come close to the complexity of reality. This paper surveys the concepts used in computing global illumination and shows examples using an experimental rendering system.



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RADIOSITY

The Radiosity method attempts to create an accurate representation of complex diffuse light environments. Once the technique is put into practice, the Radiosity method allows a new level of realism in computer graphics, including subtle effects such as colour bleeding from one object to another and light scattering into shadow regions.

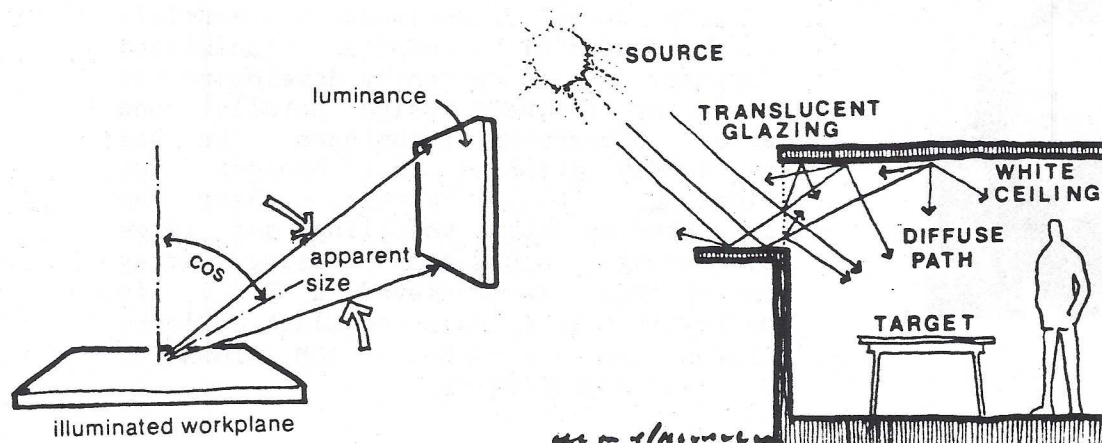
Most computer graphics shading algorithms use models that compute the sum of ambient, directional diffuse, and directional specular lighting components. The most significant additional realistic effect that the Radiosity method accounts for is the ambient term. Previously ambient light was ignored in computer graphics, or treated as a global constant.

AMBIENT LIGHT

We are surrounded by ambient light most of the time. Ambient light generally refers to light that has bounced around the scene enough times to have no obvious source or direction. Often it seems to come from everywhere, perhaps the air itself, or as the 17th Century Dutch physicist Huygens suggested, a "luminiferous ether" [9].

Ambient light is soft and subtle. It likes shadows more than directly lit surfaces, but is very difficult to pin down to a specific quantity or orientation. Most computer graphics software (until the Radiosity method) simply put all ambient light into one quantity, such as say 0.2. However nice 0.2 may be, it lacks variation if you examine it too closely. The principle motivation of the Radiosity method is to find a better approximation to this ambient value that varies locally from pixel to pixel.

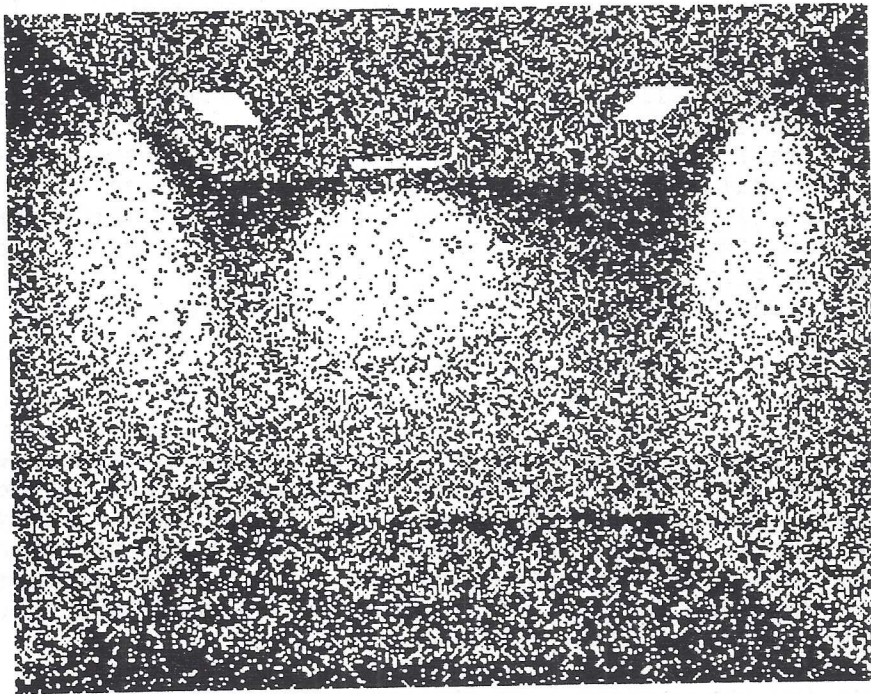
In scenes with an obvious light source such as the sun, a sky light, or a light fixture, most of the ambient light comes from the first indirect bounce of light from nearby surfaces, as shown in figure 1 (taken from [10]). In the simplest case, each surface can be considered an apparent diffuse light source.



HISTORICAL ROOTS

Research using computers to evaluate global illumination began in the late 1950s, as extensions to thermal energy transfer by architects and lighting engineers with the use of finite element techniques [14]. The application of these techniques to 3-D computer graphics began with a pioneering paper by Cindy Goral at Cornell University in 1984 [5]. This was the first time that a radiosity model was coupled directly with a 3-D hidden surface renderer, and showed the effects of a global diffuse illumination model including colour bleeding. This was followed by significant extensions to allow complex geometry and shadows [11,12,13].

Parallel developments occurred in the field of illumination engineering, where very efficient methods to evaluate global illumination were developed for restricted geometries [3,4]. There are several lighting design and analysis programs available commercially that produce perspective images. An example of such a program, L2, is shown in figure 2, courtesy of Greg Hauser, Lighting and Daylighting Analysis and Design.



L2

Sample Perspective Picture

EXPLICIT INTERREFLECTION

In the last three years the work at Cornell has progressed significantly; techniques are now developed for specular surfaces, preprocessing to allow animated scenes with moving objects, and volume densities such as smoke or dust [1,6]. Most of the work from Cornell is based on discrete polygonal patches, with the form factors computed using a large global matrix reduction. The geometric (hidden surface) relations for each patch to all other patches must be derived, and a matrix built specifying the interrelations. Various methods are used

to evaluate the patch relations, such as the hemi-cube [6], or spherical harmonic series [4]. The matrix operations can be very time consuming for large scenes. Once the matrix is solved, it contains the global illumination distribution. This has the advantage that much of the expensive computation can be performed only once, then the scene can be rendered from different points of view or animated without great expense [1]. Several spectacular images have been produced with this technique, some needing as much as 8 days of mainframe computer time.

The advantages of being able to re-compute the scene at a new view point using the same global Radiosity information has been exploited by a new high performance work station. At the SIGGRAPH-87 conference exhibition, a Hewlett-Packard display system showed a complex interior office scene with diffuse lighting. The viewer could move around the scene interactively using dials. The entire diffuse scene was redisplayed every second.

IMPLICIT INTERREFLECTION, THE RENDERING EQUATION

In a new development, James Kajiya of CalTech published a theory of rendering 3-D scenes that encompassed most known rendering techniques including radiosity. His 1986 SIGGRAPH paper refers to one 'Rendering Equation', where all previous shading methods are approximations of one kind or another to the radiant flux transfer equation [8]. As the equation itself cannot be solved by analytic means (the equation is a Fredholm Integral Equation of the second kind [14]) Kajiya proposed a Monte Carlo technique to follow Markov statistical paths through the integrand functions of the equation. This is very similar to a technique developed a year earlier for simulating the light output from fluorescent lamps [15]. The evaluation of these paths can be point samples, hence it can be implemented with distributed ray tracing methods. A major advantage of this method is that it is not dependent on a finite number of discrete patches. It evaluates the effects of global illumination at each pixel, and can work with any type of surface description for which a ray-surface intersection method exists. It also concentrates the effort on that portion of the scene that is visible to the viewer, and spends less time evaluating the hidden parts. This is potentially a very large efficiency over evaluating all light everywhere in the scene as the Radiosity method requires. However, due the reliance on ray tracing, this new method does not exploit spatial or temporal coherence.

The second important feature of the 'Rendering Equation' method is that it easily combines specular reflection effects, unlike the Radiosity method which is restricted to diffuse interreflection. This disparity has recently been rationalized at Cornell by a hybrid technique that combines both the Radiosity method and Ray tracing in 2 passes [16].

RADIOSITY AT AMAZING ARRAY PRODUCTIONS

An experimental rendering system has been constructed by the author that evaluates diffuse illumination using the rendering equation method. The rendering equation method uses discrete point sampling

and traverses an iterative Markov chain to evaluate the integral equation directly, without reducing the model to a finite number of polygonal patches. In our implementation of this idea, we use stratified Poisson distribution sampling, rather than heirarchical sampling mentioned by Kajiya [8]. We also make use of importance sampling methods, encoding prior geometric information into the sampling system and varying the granularity of the integral at higher terms. We have derived an algorithm for evaluating the rendering equation with a uniform data structure, without recursion, in such a way as to exploit the vector processing facility of the NMX-432 array processors. The techniques for vectorizing rendering computations on array processors were developed over several years for use in molecular modelling [2].

Several examples of the diffuse illumination algorithm are shown at the end of the paper. The example with one sphere compares the difference between ordinary ray tracing and the rendering equation technique. An additional example shows the effects of radiant object surfaces.

COMPUTING RESOURCES

The diffuse illumination algorithm at Amazing Array Productions is implemented using a parallel computing system consisting of heterogeneous processing elements. The system is composed of an ELXSI 6400 scientific computer with four CPU elements, 128 Mbytes of main memory, and 2.4 Gbytes of disk storage. The ELXSI runs a message based operating system (EMBOS) that dynamically distributes processing load among available CPUs. Most of the intensive floating point computations are performed using four Numerix NMX-432 array processors. Each array processor has a parallel-pipelined architecture, and is rated by the manufacturer at 30 MFLOPS (million floating point operations per second).

TOY INDIRECT LIGHT EVALUATOR

The following pseudo code represents the outline of a simple method to start experimenting with Radiosity. In this form it makes many severe assumptions, and does not easily account for the total global illumination. It outlines the beginning of adding Radiosity type effects into an existing ray-tracing system, with very little extra code. Note that this does not account for multiple diffuse interreflections, or colour bleeding, and is at least 100 times more expensive than normal ray tracing.


```

FOR (each pixel(x,y) in the image ) {
  cast ray from eye through pixel (x,y) into scene
  test ray with all objects, find nearest intersection
  set Radiation to zero
  FOR (a fixed number of iterations, say 100) {
    choose random ray direction in 3-D space
    IF (ray direction > 90 degrees from surface normal) {
      reverse ray direction
    }
    fire new ray in this direction with base at visible surface
    test ray with all objects, find nearest intersection
    temporarily evaluate shade at this new intersection
    Radiation = Radiation +
      (temporary new shade) / (iteration count)
  }
  shade visible surface scaled by Radiation
  display pixel (x,y)
}

```

CONCLUSION

Taking global diffuse illumination into account while rendering computer graphics images significantly enhances realism. Several methods are available: the Radiosity method finds a global solution independent of view position, the rendering equation method is view dependent but can render a broader class of objects and easily evaluate specular reflections, transparency, refraction, motion blur, and translucency, and does not require objects to be subdivided into patches. A new class of hybrid algorithms combines the best features of each. There are many areas of active research, including techniques for improving computational efficiency. It is hoped that this survey stimulates more interest in this powerful technique.

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